

search query at the same time, each server processes its own chunk of data one piece at a time.

While distributed computing can accelerate searches, even the most powerful systems face limitations when it comes to processing vast amounts of data, especially as databases continue to grow exponentially. This is where new computing methods, such as quantum computing, are poised to bring about a transformative difference. Instead of processing tasks sequentially, a quantum computer can evaluate all possible solutions at once. This capability comes from a key concept in quantum mechanics known as superposition. So, instead of checking each result one by one or relying on distributed computing, a quantum computer can examine all possibilities simultaneously, delivering the solution much faster.

Let's revisit the Amazon search example. Currently, Amazon doesn't use quantum computing; instead, it relies on highly optimised classical computing systems with distributed computing to quickly sort through millions of titles and return relevant results. But imagine if Amazon were to adopt quantum computing in the future. With a quantum computer, all possibilities could be evaluated at the same time, drastically reducing the time required. What might take minutes or even hours on a traditional system could potentially be completed in seconds with quantum technology.

This potential isn't just theoretical—it holds immense real-world implications. For instance, when you search for information on Google, classical computers use sophisticated algorithms and distributed processing to evaluate and rank results. With quantum computing, however, such systems could analyse all possible results at once, potentially returning the most relevant answers in a fraction of a second, transforming the speed and accuracy of search as we know it.

The qubit advantage

Quantum computing represents a fundamental shift from traditional computing. While classical computers process data using bits, which are either 0 or 1, quantum computers rely on qubits. A qubit, or quantum bit, is unique because it can exist in both 0 and 1 states simultaneously, thanks to a phenomenon known as superposition. This superposition means that each qubit added doubles the number of possible states. For example, with a single qubit, there are only two possible states: 0 or 1. However, with two qubits, there are four possible states: 00, 01, 10, and 11. By the time we reach three qubits, the states increase to eight combinations (000, 001, 010, 011, 100, 101, 110, and 111). In general, n qubits can represent 2^n states. Therefore, four qubits can represent 16 different states simultaneously, and adding even one more qubit doubles this to 32 states. This exponential growth provides quantum computers with an unprecedented level of computational capacity, enabling them to tackle complex problems that are beyond the reach of classical systems.

Looking towards the future, as quantum computing progresses, we could see systems with hundreds or even thousands of qubits, potentially reaching a power far beyond today's classical systems. For instance, a quantum computer with 100 qubits could represent 2^{100} states, an astronomically large number, enabling it to solve problems that would be impossible for classical computers, even with millions of processors.

However, qubits are highly sensitive and can be disrupted by environmental factors such as heat, electromagnetic radiation, or even slight vibrations. These disturbances cause 'quantum decoherence' where the qubit loses its quantum state and the system collapses into classical behaviour. To prevent this, quantum computers typically operate at extremely low temperatures, close

to absolute zero, to minimise thermal fluctuations. Specialised cooling techniques, like dilution refrigerators, are used to lower the temperature of qubits to just a fraction of a degree above absolute zero. Additionally, the materials and design of the quantum processor must be carefully engineered to shield qubits from external noise.

Superposition

In classical computing, a byte (which is made up of 8 bits) can store values like 1, 0, or even a combination like 10101100. But a qubit works differently. It can represent 0, 1, or both 0 and 1 at the same time until it is measured. You could think of it as having a value of 0.5 for 0 and 0.5 for 1, meaning there is an equal chance of it being 0 or 1 when checked, instead of just being one or the other. This ability of qubits to exist in multiple states simultaneously is what gives quantum computers their power. While a regular computer must process each possibility one by one, a quantum computer can process many possibilities at once.

Imagine you're looking for a specific movie on a streaming platform like Netflix. In traditional computing, a classical computer would need to go through the list of movies one by one, checking each to see if it matches your search. With a quantum computer using superposition, you can look at the entire library of movies at once, all in parallel. Instead of checking each movie one by one, a quantum computer can process all possible matches at the same time. This means that you can find the movie you are searching for in a fraction of a second, instead of waiting minutes or even hours to scroll through the results.

Here is another example. Let's say you're looking for a particular product on Amazon, but there are millions of items in the store. A classical computer would search through the items one by one, or at best, in parallel chunks, but still check each item sequentially. A quantum computer can search through